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Time-to-event analysis for hospital stays caused by respiratory infections

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Abstract

Background: Time-to-event data from hospitalized patients with five different respiratory infections, listed as Influenza A, RSV, Influenza B, and SARS-CoV-2 and Omicron.

Methods: Gender, age, temperature, whether or not they had a respiratory infection in the hospital, pre-existing conditions, treatments, symptoms, and four laboratory values, were collated for 2251 patients at the beginning of their hospitalization. The effect of these factors on the duration of hospitalization due to the five respiratory infections was analyzed by using Cox regression. After using multiple imputations, multifactor analysis, and correlation tests were performed. Also by restricted mean survival time (RMST) to compare the effect of each variable on the length of hospitalization. All results were calculated by RStudio software where statistical significance was set as $p < 0.05$. For multivariate Cox models, Martingale residuals to assess the functional form for covariates and scaled Schoenfeld residuals detect departures from proportional hazards.

Results: The result is based on Cox regression and RMST, in this thesis survival time is hospital stays, survival event is normal discharge, and the event associated with censored data is patient death. The effect of each factor on the hospital stays was compared by comparing Hazard Ratio (HR) and 95% confidence intervals (CI). The effect of nosocomial infection on hospital stays was significant for all five respiratory infections(except for SARS-Cov-2 HR=0.52, other four HR<0.5). Age had a significant effect on Omicron and Influenza A (each p<0.001) when age was divided into two groups using a cut-off of median age. The hospital stays were longer for those older than the cut-off than for the younger group (each HR<0.5). The probability of discharge was reduced if the patient had a bacterial infection (each HR<0.6, p<0.001). The hospital stays were also increased in patients who also had Diabetes beforehand in SARS-CoV-2 patients (HR=0.67, 95% CI 0.49-0.91), RSV patients (HR=0.77, 95% CI 0.60-0.98), and Omikron patients (HR=0.77, 95% CI 0.60-0.98). Cardiac Problems Have increased the length of Hospitalization in SARS-CoV-2 Patients (HR=0.63, 95% CI 0.48-0.82) and in Omikron Patients (HR=0.77, 95% CI 0.63-0.92). Dialysis-Renal Insufficiency affects increasing length of stay in SARS-CoV-2 patients (HR=0.61, 95% CI 0.44-0.84), Omikron patients (HR=0.75, 95% CI 0.59-0.93), Influenza A patients (HR=0.66, 95% CI 0.54-0.81), and Influenza B (HR=0.65, 95% CI 0.49-0.85). Laboratory values are also associated with hospital stays. However, for each unit increase in LDH, the probability of discharge increased for RSV patients (HR=1.07, 95% CI 1.01-1.144) and for Omicron patients (HR=1.12, 95% CI 1.07-1.16). RMST analyzed selected factors in the Omicron group compared to the SARS-CoV-2 group.

RMST of up to 30 days for SARS-CoV-2 patients with nosocomial infection was about 21.78 days, and those without was about 15.12 days. For Omicron group, the result is 23.29 days compared with approximately 9.52 days for those with non-nosocomial infections. RMST results show nosocomial infections, bacterial infections, and pre-existing conditions can prolong hospitalization. Four of the five groups indicated a higher correlation between CRP and bacte-

rial infection ($0.3 < \text{cor} < 0.4$), out of the five groups, diabetes was found to be associated with blood glucose levels in two of them($0.3 < \text{cor} < 0.4$).
Conclusions: With the Cox regression method, all factors decreased the probability of discharge except for elevated LDH per unit, which increased the probability of discharge; in other words, the presence of most factors either increased hospital stays. The trend of the results of the RMST method is consistent with the Cox regression method.

Keywords: Influenza A; RSV; Influenza B; SARS-Cov-2; Omicron; Cox regression; RMST

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List of Abbreviations

- SARS-COV-2...Severe Acute Respiratory Syndrome Coronavirus 2
- RSV...Respiratory syncytial virus
- COPD...Chronic Obstructive Pulmonary Disease
- LDH...Lactate dehydrogenase
- CRP...C-reaction Protein
- Leuko...Leukocyte
- nosocomial...nosocomial infection
- HR...Hazard Ratio
- KM...Kaplan-Meier
- cor...correlation coefficient
- RMST...restricted mean survival time
- DRI...Dialysis-Renal Insufficiency
- tau...truncation time

1 Introduction

Before the COVID-19 pandemic, the most common viruses of respiratory infections mainly included respiratory syncytial (RSV) and influenza A/B. Respiratory infections caused by Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) dominated after the COVID-19 pandemic began in early 2020. Since 2022, Omicron, a variant of SARS-CoV-2 has assumed responsibility for COVID-19[1]. Differentiating these five respiratory illnesses is difficult, especially during influenza season, because common clinical appearances of COVID-19, including fever, cough, and myalgia or fatigue, mimic those of influenza A/B and RSV[2][3][4]. Therefore the diagnosis of respiratory infections cannot be based on symptoms only.

Since the beginning of the pandemic, multiple studies have explored the association between pre-existing conditions and respiratory illness outcomes[5]. Nosocomial infections (NIs) and bacterial infections are also being explored[6][7]. Nosocomial infections (NIs) are usually described as infections acquired during hospitalization within 48–72h after admission[8]. Some studies have shown that bacterial infections have also been linked to the influenza virus, leading to worsening of the patient's symptoms and even death[9]. C-reactive protein (CRP), leukocytes (leuko), lactate dehydrogenase (LDH), and glucose are commonly used as markers to assess the severity of various diseases and inflammatory conditions.

Both RMST (Restricted Mean Survival Time) and Cox regression are methods used in time-to-event analysis. Cox proportional hazards regression provides an effect estimate by quantifying the difference in survival between patient groups, and the effect estimate of Cox regression analyses is a hazard ratio (HR), which is the ratio of two hazards. The hazard is the incidence rate of an event in an infinitesimally short time period[10], in this thesis refers to discharge. RMST is the mean survival time of all subjects in the study population followed up to truncation time (tau) and is simply the area under the survival curve up to tau[11].

The thesis aims to compare the hospital stays of patients with five respiratory infections using time-to-event analysis and assess the influence of multiple factors on the hospital stays and underline the differences and similarities between five respiratory illnesses, and a comparison is made between Cox regression and RMST.

2 Materials and Methods

2.1 Patients

In this thesis, a total of 2114 cases tested positive for Influenza A (n=682), RSV (n=336), Influenza B (n=306), SARS-CoV-2 (n=318), and Omicron (n=472) in six high seasons for the respiratory infections in sequential years were evaluated after removing unavailable data.
The proportion of the 2114 valid records that were successfully discharged without death was 89.8% (Table 1), of which the number of hospital stays of more than one week (including 7 days) was 1342. The amount of censored data in this thesis is the death record. With the exception of SARS-CoV-2, which had a censored rate of more than twenty percent at 23.9%, the censored rates of the other four groups were kept below ten percent. Meanwhile, approximately 66,6% of the patients were older than 70 (including 70).

Until season five, the viruses causing respiratory infections were influenza A/B and RSV. All

Virus (n)	Influenza A 682	RSV 336	Influenza B 306	Sars-Cov-2 318	Omicron 472
Normal discharges	630	304	283	242	440
censored data death	52	32	23	76	32

Table 1: Distribution of patient populations in the dataset

recorded hospitalization data during 5. season was from respiratory infections caused by SARS-CoV-2. Since Omicron, a variant of SARS-CoV-2 has assumed responsibility for COVID-19, the previously common RSV and Influenza A/B have resurfaced in hospitalization records (Table 2). Before the two COVID-19 viruses were recorded, i.e., the first four seasons, the first, third, and fourth seasons were the peak of influenza A infections, between 61% and 71%. The second season, in contrast, was the peak for Influenza B. RSV infections fluctuated between 12%-37% in the previous four seasons (Figure 1).

	Influenza A	RSV	Influenza B	Sars-Cov-2	Omicron
season 1	190	98	25	0	0
season 2	83	49	282	0	0
season 3	187	109	0	0	0
season 4	123	43	7	0	0
season 5	0	0	0	339	0
season 6	152	64	0	0	500

Table 2: The number of patients with five respiratory infections recorded in six high seasons for the respiratory infections in sequential years

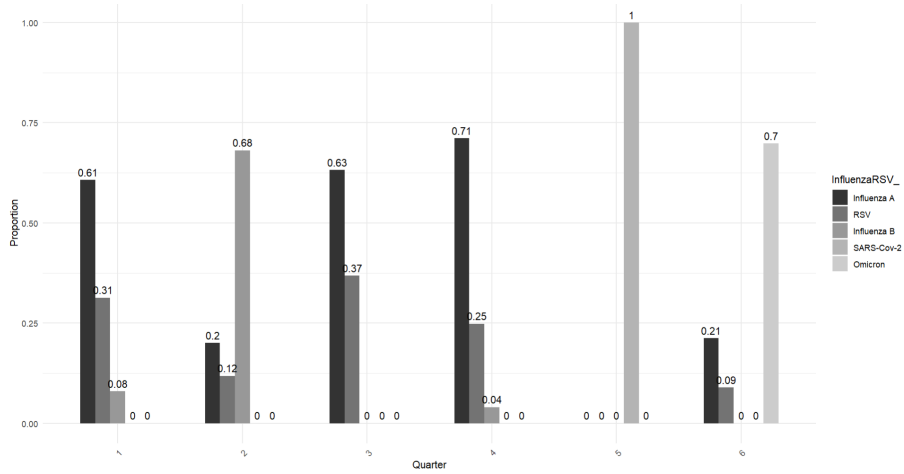

Figure 1: Proportion of people in six high seasons for respiratory infections in sequential years (from darkest to lightest color, Influenza A, RSV, Influenza B, SARS-CoV-2, and Omicron)

Patients infected with RSV had the highest average age of 75.6 years, which was generally the oldest, and those infected with influenza B had an average age close to that of the RSV group, which was 75.3. In contrast, the SARS-CoV-2 group had the lowest average age, 71.9 years. The average age of the Influenza A and Omicron groups was close to each other, 72.7 and 73.4 years (Figure 2). In each group, the number of males and females was relatively similar. Overall, the number of men infected with respiratory infections is higher than women. Only the Influenza A group had slightly more fever than no fever at the time of hospitalization enrollment (Table 3).

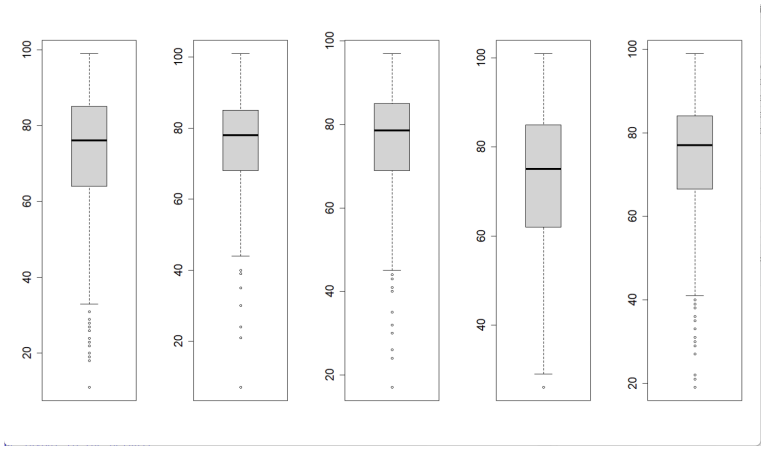

Figure 2: The age distribution, from left to right: InfluenzaA, RSV, InfluenzaB, SARS-CoV-2, and Omicron

The analyzed dataset was from patients with respiratory infections who tested positive for Res-

Virus(n) (n)	Influenza A 682	RSV 336	Influenza B 306	Sars-Cov-2 318	Omicron 472
Age	72.7	75.6	75.2	71.9	73.4
Gender (m/f)	378/304	193/143	167/139	161/157	243/230
Temperature (°C)	275/320	157/130	146/128	144/117	197/158

Table 3: Demographic parameters of five respiratory infections: average age; Temperature < 37.2°C /Temperature >= 37.2°C

piratory syncytial virus (RSV), Influenza A/B and SARS-CoV-2, and Omicron. Meanwhile, the following data were assessed to analyze the impact of relevant factors on the hospital stays: age, gender, nosocomial infection, body temperature, bacterial infections, laboratory results (LDH, CRP, leuko, glucose), and pre-existing conditions (Chronic Obstructive Pulmonary Disease, Diabetes Mellitus, Tumor, Cardiac problem, Dialysis-Renal Insufficiency) upon patients admission. The study used the time-to-event method to analyze the impact of the above factors on the hospital stays. The following data were also recorded: symptoms (Cough, dyspnea, headache, fatigue, pneumonia, diarrhea), treatments (Antibiotika, Tamiflu, and ventilation). Nevertheless, due to the time-to-event property and the uncertainty of when treatments (e.g., Tamiflu, antibiotics) and symptoms (e.g., cough) began and ended, the analysis focused on the patient's case data at the time of admission. The study does not consider the effect of recorded factors on patient mortality and is only interested in the length of hospitalization in cases with normal discharge.

2.2 Statistical Analysis

The study viewed hospital stays as survival time. The event is normal discharge, not including death records. Death data were censored data in this study.
The original data were divided into five groups according to the type of respiratory infection, the age of the patients was divided into two groups according to the median of each group, and 37.2 was used as a cut-off to divide the temperatures into two groups. The human body typically has established reference ranges for four laboratory values, when the body is inflamed, this value can soar to tens or even hundreds of times its original value, so the use of normalization on the detected values makes the values (log CRP, log LDH, log Leuko, log Glucose) not overly affect the results. Normalization also removes the effect of units, since laboratory values have different units of measure, making their effects in the model comparable.
RMST and Cox-Regression are both methods for analyzing survival time, but they have different

emphases. Cox regression is a semiparametric regression method that is based on HR, therefore its results show the association of each covariate with survival risk while not directly giving the absolute value of survival time.

RMST is a non-parametric method that estimates the mean survival time up to a specific time point, and it requires complete data for accurate and unbiased estimations. Therefore for RMST analyses, handled missing values by using multiple imputations, the random forest (rf) of imputations is used. RMST calculates the mean value of survival time at a specific cut-off time tau (Figure 3).

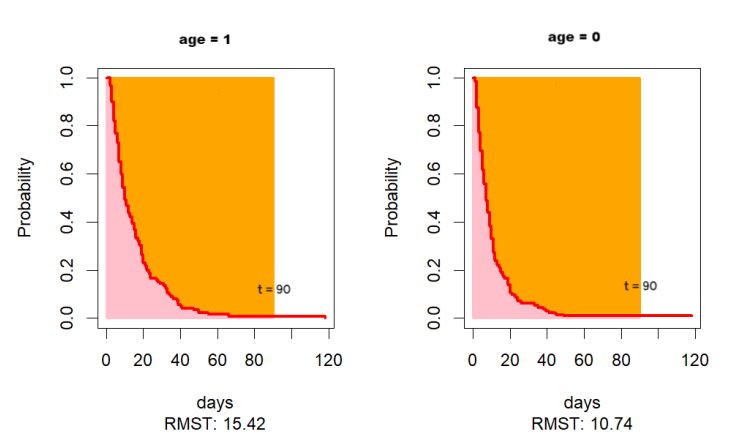

Figure 3: Example of survival curves for age (median) versus length of patient hospitalization for both groups, age=1 indicates older than median age. tau=90 means that the analysis has been conducted only on data from the last 90 days.

RMST provides a direct measure of survival time. RMST is used to compare differences in survival time between different groups of the same covariate.

There are two basic assumptions for the Cox Proportional Hazards model. The first is the assumption of proportional hazards. This means that the HR for any two individual factors does not depend on time. The second assumption is that of the functional form of the covariates[12]. Covariates that do not change regularly over time are said to satisfy the proportional hazards assumption. The proportional hazards assumption is generally tested using the Schoenfeld test, when $p>0.05$, means that there is no significant change in the relationship between the risks over time, i.e., the factors fit the proportional risk assumption. Martingale residuals can help determine whether the model satisfies the independence assumption, i.e., whether the model's residuals are independent in time. Checking the Cox model with cumulative sums of martingale-based residuals[13]. Cox-Snell Residuals is a graphical scale for evaluating the fitness of Proportional hazard models; the short deviation of residuals from the straight line through the origin with a slope of 1, the more appropriate fitness of the survival model[14][15].

All statistical analysis was conducted using R, version 4.1.2.

3 Results

3.1 Comparison of five respiratory infections by Cox regression

Univariate analysis was first done for each of the five groups.

3.1.1 General situation and laboratory parameters

After Cox regression analysis it can be concluded that gender and temperature do not have a significant effect on the hospital stays.
Influenza A had the shortest average hospital stays (10.15 days), while the SARS-CoV-2 group had the longest average (17.66 days) hospital stays. The average hospital stays for the remaining three respiratory infections is as follows: 12.22 days for RSV, 8.68 days for Influenza B, and 12.32 days for Omikron. From the box plots, it can be seen that Influenza A and RSV have more extreme values for hospital stays (Figure 4).

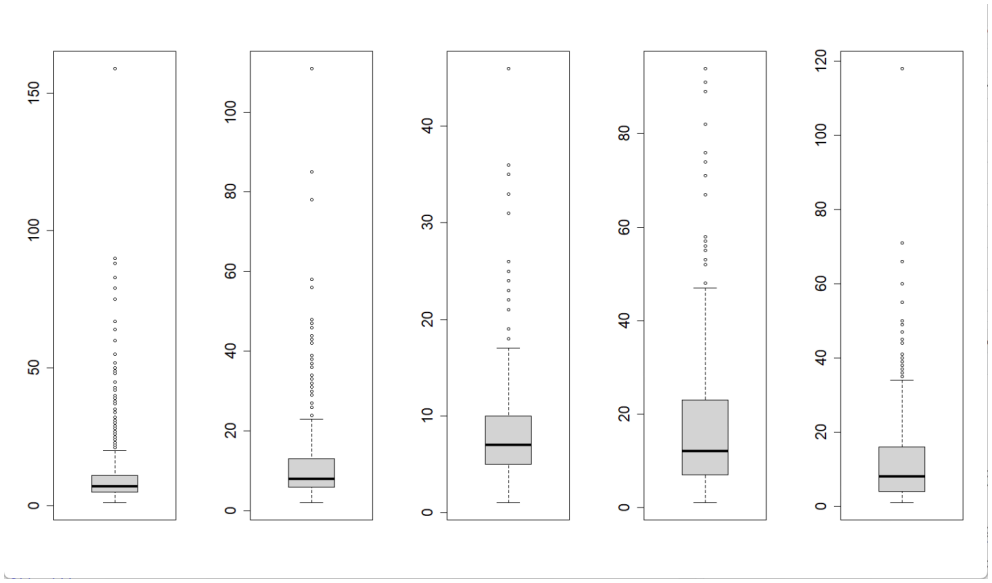

Figure 4: Box plots of the duration of hospitalization for five respiratory infections, From left to right, Influenza A, RSV, Influenza B, Sars-Cov-2, and Omicron.

Age was divided into two groups according to the median. According to the results of Cox regression, the age factor had a statistically significant effect on the Influenza A and Omicron groups. The Influenza A age group (<76) was compared to those 76+. The exp(coef)/HR for the 76+ group was 0.76 (Table 4). The event of the Cox regression model was discharged, when the HR was 0.76, indicating that the probability of discharge for patients aged more than or equal to 76 years was 76% of the probability for patients aged less than 76 years. Based on this situation, it appeared that individuals aged 76 years and above experienced an increase in hospital stays

compared to those under 76 years old when all other factors being equal. In Omicron group, age 77+ was used as the reference group (age < 77), the older group experienced an increase in hospital stays compared to individuals aged less than 77. The age factor had a greater impact on the Omicron group, the more age, the lower the probability of being discharged. The probability of discharge for Influenza A patients also decreased with age, and hospital stays increased but were not as greatly affected as for Omicron patients (Figure 5).

Virus	HR	p
Influenza A 95% CI	0.77 (0.65, 0.89)	p<0.001
Omicron 95% CI	0.66 (0.55, 0.80)	p<0.001

Table 4: time-to-event result: Age

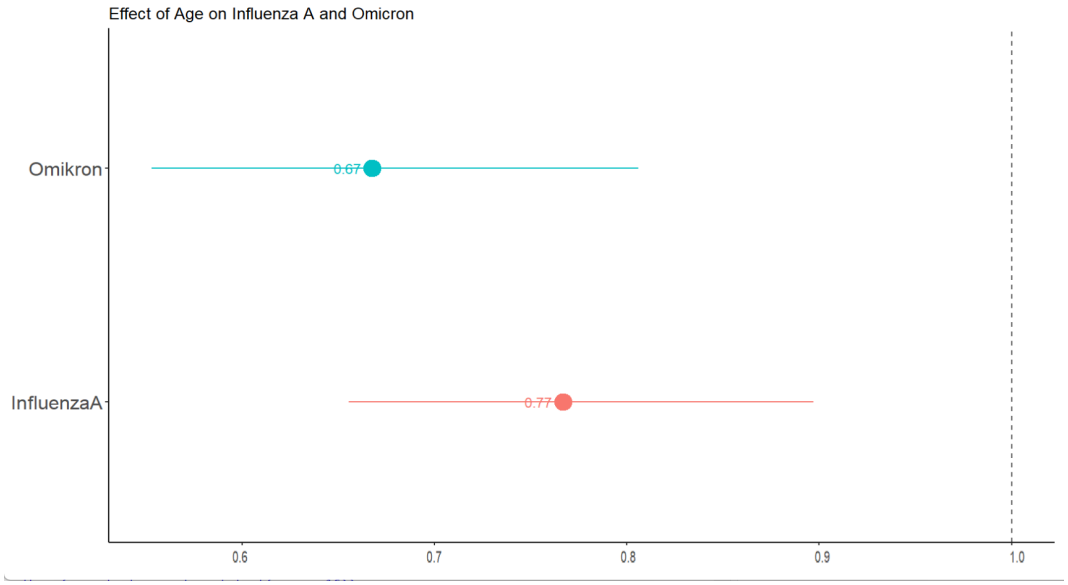

Figure 5: Effect of Age on Influenza A and Omicron groups, HR values from top to bottom are: 0.67, 0.77

The table below records the average laboratory values for the five groups (Table 5). Among all the infections, RSV group had the highest leukocyte count (10.25 n/ul) and the Influenza B group had the lowest leukocyte count (7.40 n/ul). The counts of the other three groups were approximately at the same level.

The group infected with SARS-CoV-2 showed the highest level of CRP (83.45 mg/dL), while the Influenza B group exhibited a lower level (48.58 mg/dL). These findings indicate that CRP is much higher in patients infected with SARS-CoV-2.

Lactate dehydrogenase level (373.57 IU/mL) was highest in SARS-CoV-2 group and lowest in

Omicron group (247.26 IU/mL). LDH is much higher in the bodies of patients infected with SARS-CoV-2.
RSV group had the highest blood glucose level (144.63 mg/dL) and the Omicron group had the lowest level (126.55 mg/dL).
From the analysis of these data, obvious differences in laboratory values were observed between the different respiratory infections. In particular, the SARS-CoV-2 group had the highest CRP and LDH levels, while the Influenza B group had the lowest leukocyte counts. The RSV group had higher leukocyte and blood glucose counts.

Virus(n)	Influenza A	RSV	Influenza B	SARS-CoV-2	Omicron
Leukocytes (n/ul)	8.63	10.25	7.40	8.52	8.28
CRP (mg/dL)	67.67	64.08	48.58	83.45	53.77
LDH (IU/mL)	307.41	264.86	273.57	373.57	247.26
Glucose (mg/dL)	137.40	144.63	135.15	128.44	126.55

Table 5: Average of four laboratory values of five respiratory infections

After conducting Cox regression analysis, the impact of normalized laboratory values on five groups was studied. Specifically, laboratory values that had a statistically significant effect on hospital stays were gathered and analyzed (Figure 6).
The trend for the impact of laboratory values on hospital stays for different respiratory infections was the same, i.e., the trend of increasing or decreasing the risk of survival time (Hazard) per unit increase in laboratory values is not different for different viruses, except for LDH. An increase in CRP, blood glucose, and Leukocyte count per unit means an increase in hospital stay. The impact of higher blood glucose on the hospital stays is most noticeable.
The effect of laboratory values on hospital stays for different respiratory infections was more consistently presented on Influenza A and Influenza B. Both CRP and Leukocyte increase lengthen the hospital stay in patients with these two respiratory infections by about the same multiple. High blood glucose levels impact the duration of hospitalization for Influenza A patients, increasing the length of stay (HR=0.75), but it is not significant for Influenza B patients.
Omicron is a variant of SARS-CoV-2, it is meaningful to compare them. For SARS-CoV-2, higher levels of CRP and glucose were associated with longer hospital stays (HR=0.79, HR=0.41, respectively), while for the Omicron patients, higher levels of lactate dehydrogenase were associated with shorter hospital stays. Both viruses exhibited similar trends in the association of C-reactive protein with hospital stays, with higher levels being linked to longer stays, the HR values are all around 0.80.

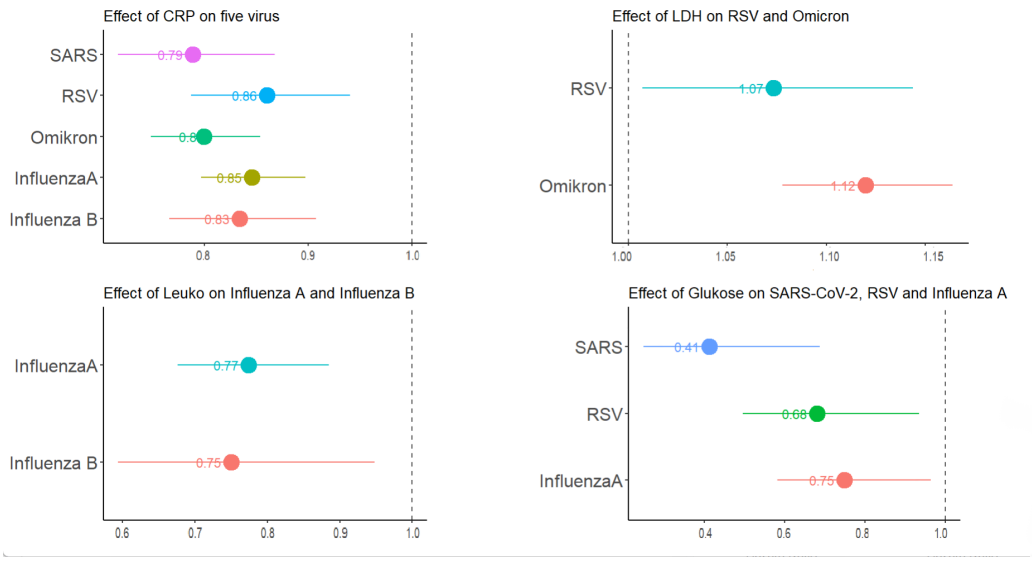

Figure 6: Effect of normalized laboratory values on the five groups upon Cox regression analysis, the positions of the dots in the figure are the HR values, and the line extension is the 95% CI. HR values from top to bottom are: 0.79, 0.86, 0.80, 0.85, 0.83 (top left)/ 1.07, 1.12 (top right)/ 0.77, 0.75 (bottom left)/ 0.41, 0.68, 0.75 (bottom right)

3.1.2 Pre-existing conditions

The pre-existing conditions that were recorded were Chronic Obstructive Pulmonary Disease (COPD), Diabetes Mellitus, Tumor, Cardiac, and Dialysis-Renal Insufficiency (DRI). It has been consistently observed that pre-existing conditions tend to lengthen hospital stays (Figure 7). However, the number of people with only Diabetes Mellitus or DRI is very limited, accounting for less than 0.1% of the patients (Figure 8-9).

In the context of Influenza A patients, the presence of Tumors and COPD exhibited significant associated with the probability of hospital discharge, with HR of 0.73 and 0.78, respectively. Patients with both Diabetes Mellitus and RSV or SARS-CoV-2 or Omicron respiratory infections had 67%, 77%, and 77% less likely to be discharged, respectively. The presence of cardiac problems and DRIs concomitantly have an impact on hospital stays for patients with SARS-CoV-2 and Omicron respiratory infections, with the HR of 0.63 and 0.77, respectively.

It is notable that among the pre-existing conditions associated with SARS-CoV-2 and Omikron, the highest number of people had only heart problems (Figure 8-9).

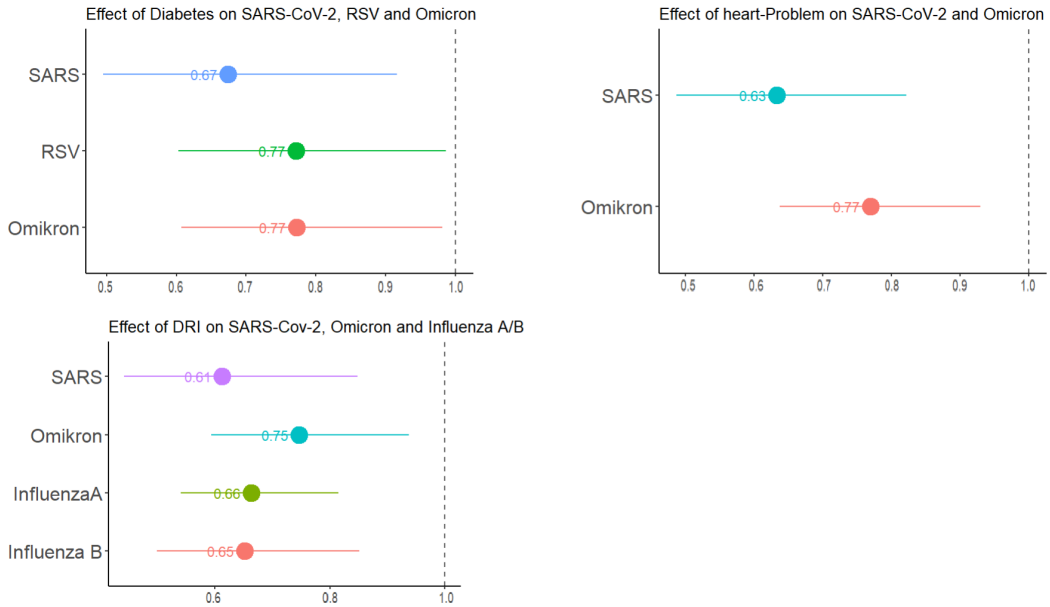

Figure 7: Effect of Pre-existing conditions on five groups, all HR values < 1, HR values from top to bottom are: 0.67, 0.77, 0.77 (top left)/ 0.63, 0.77 (top right)/ 0.61, 0.75, 0.66, 0.65 (bottom left)

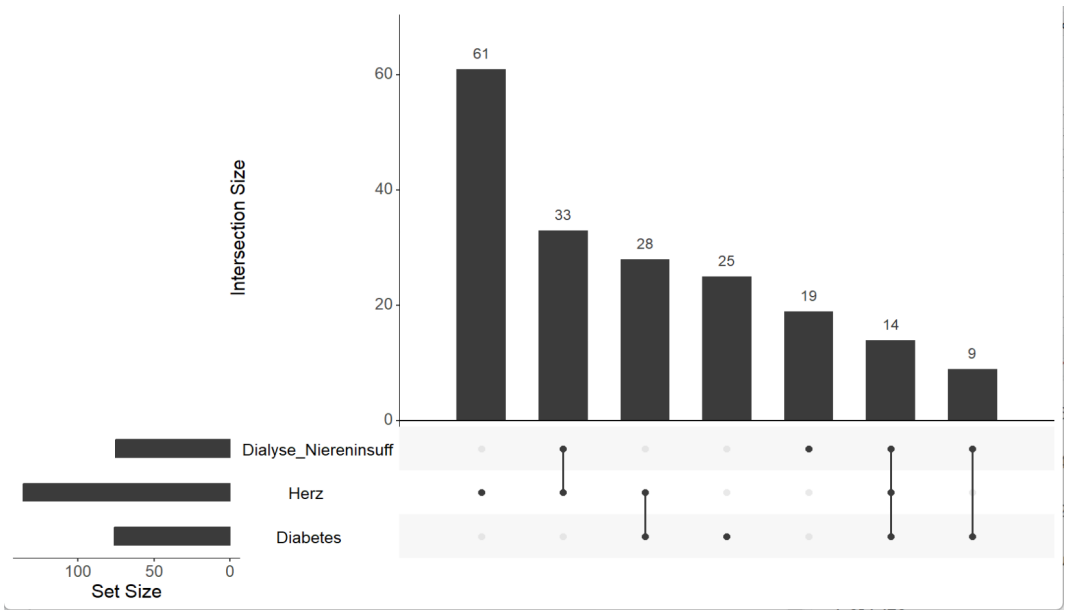

Figure 8: Pre-existing conditions in SARS-CoV-2 group, the three pre-existing conditions from top to bottom are DRI, heart problem, diabetes. Where a dot is present means that the condition exists. The numbers on the bar graph from left to right are 61, 33, 28, 25, 19, 14, and 9.

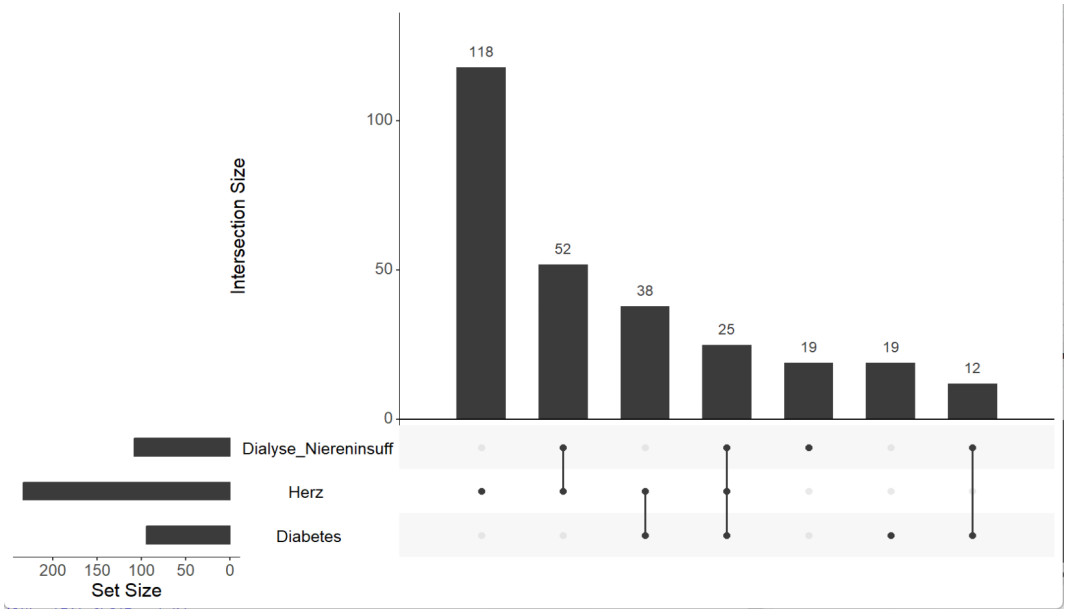

Figure 9: Pre-existing conditions in Omicron group, the three pre-existing conditions from top to bottom are DRI, heart problems, and diabetes. The numbers on the bar graph from left to right are 118, 52, 38, 25, 19, 19, and 12.

3.1.3 Nosocomial infection and bacterial infection

Among the infections studied Omicron demonstrated the strongest association with nosocomial infection, with a HR of 0.30, indicating patients with nosocomial infections had a 70 percent lower risk of hospital discharge compared to those with non-nosocomial infections, in other words, patients who were nosocomial infections have longer hospital stays. Similarly, nosocomially infected patients with RSV also exhibited significant associations with hospital stays, with HR values of 0.37. The other three viruses, Influenza A, Influenza B, and SARS-CoV-2 all had approximately 0.5 for HR, and the impact was also very pronounced (Figure 10).
Patients with respiratory and bacterial infections had longer hospital stays than those without bacterial infections, except for the Omicron group. There was no statistically significant association between the length of hospitalization and the presence or absence of bacterial infection in Omicron group. After conducting Cox regression models obtained HR for four infections: Influenza A (HR=0.57), RSV (HR=0.59), Influenza B (HR=0.46), and SARS-CoV-2 (HR=0.39) (Figure 11). In SARS-CoV-2 infections, HR of 0.39 means that patients with bacterial infections are hospitalized for significantly more time than those without bacterial infections.

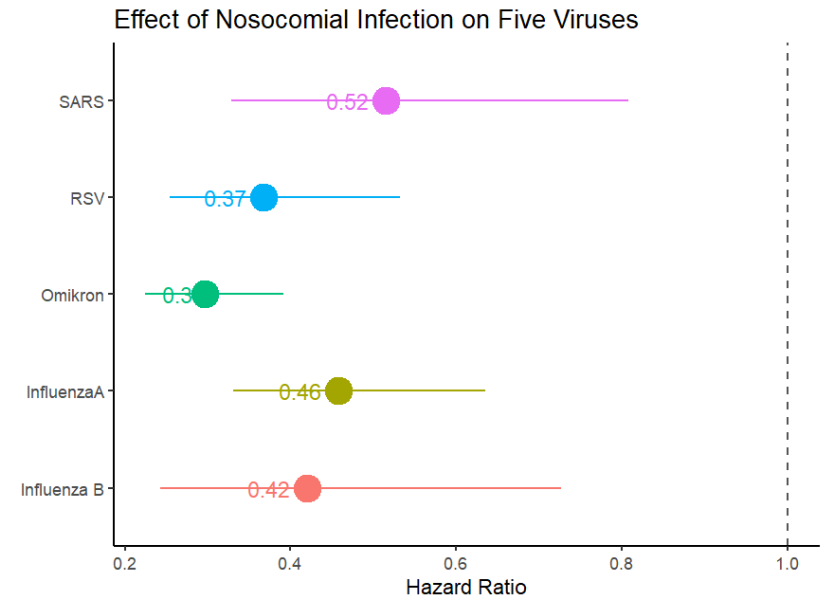

Figure 10: Comparison of the effect of the presence of nosocomial infections on the length of hospitalization, HR values from top to bottom are: 0.52, 0.37, 0.30, 0.46, 0.42

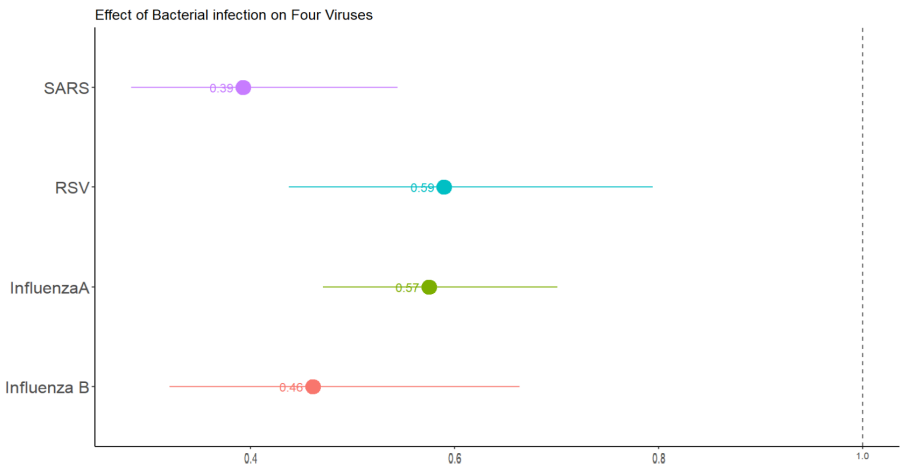

Figure 11: Comparison of the effect of the presence of bacterial infections on the length of hospitalization, HR values from top to bottom are: 0.39, 0.59, 0.57, 0.46

3.2 Correlations

All correlation analyses were performed under Multiple Imputations.

The most significant correlation in the Influenza A group was between log(CRP) and bacterial infection with a correlation coefficient (cor) of 0.33 (Figure 12). In RSV group it was a correla-

tion of diabetes and blood glucose with cor=0.4 (Figure 13). In the Influenza B group, log(CRP) and bacterial infection with cor=0.3 and log(CRP) and log(Leuko)with cor=0.38 (Figure 14). In SARS-CoV-2 group correlation coefficient of diabetes and blood glucose was 0.43. cor of log(CRP) with bacterial infection=0.35 (Figure 15). In the Omicron group was a previously unseen correlation between heart problem and age cor=0.34 (Figure 16).
Four of the five groups indicated a higher correlation between CRP and bacterial infection, out of the five groups, diabetes was found to be associated with blood glucose levels in two of them.

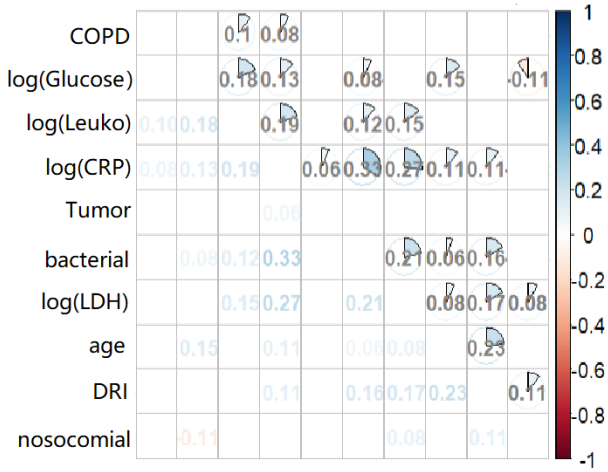

Figure 12: Correlations plot for Influenza A group. log(CRP) and bacterial infection with cor = 0.33

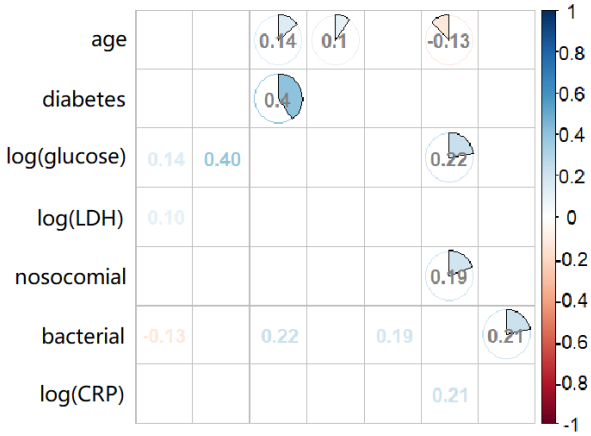

Figure 13: Correlations plot for RSV group. Diabetes and blood glucose with cor=0.4

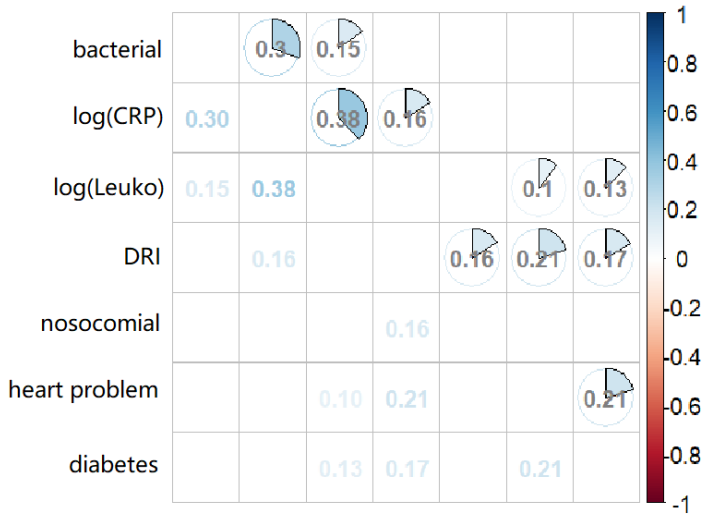

Figure 14: Correlations plot for Influenza B group. log(CRP) and bacterial infection with cor=0.3 and log(CRP) and log(Leuko)with cor=0.38

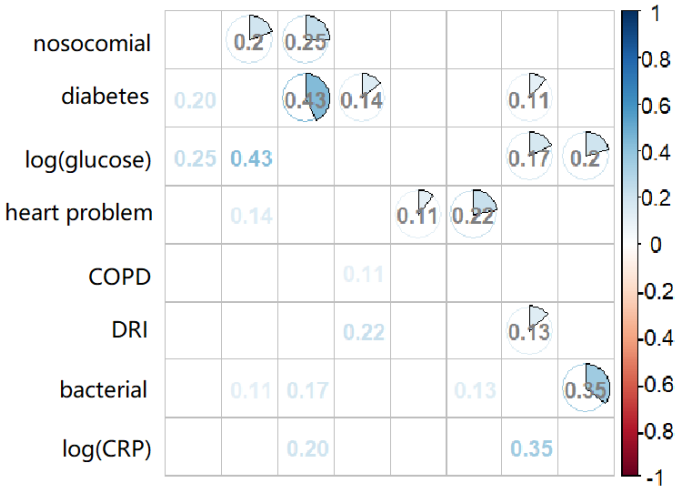

Figure 15: Correlations plot for SARS-CoV-2 group. Diabetes and blood glucose with cor=0.43

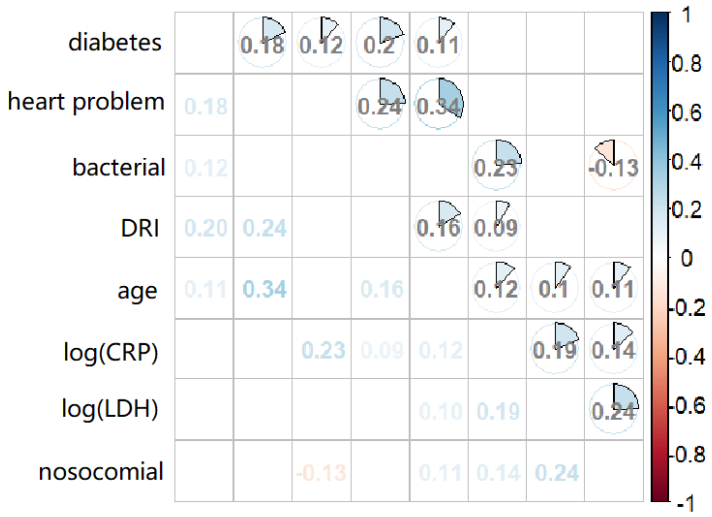

Figure 16: Correlations plot for Omicron group. Heart problem and age cor=0.34

3.3 Goodness-of-Fit Testing for the Cox Proportional Hazards Model

The factors that had a significant effect on hospital stays for each respiratory infection were put into multivariate models. The proportional hazards assumption and independence assumption were tested under the condition that the Cox model was statistically significant ($p<0.05$ for each variable).

This thesis used martingale residuals to assess the functional form for covariates and scaled Schoenfeld residuals to detect departures from proportional hazards.

3.3.1 Proportional Hazards Assumption

For Influenza A patients, the hospital stays served as the dependent variable, and log(CRP), log(Leuko), bacterial infection, COPD were the independent variables. The results of the Schoenfeld test are as Figure 17. The proportional hazards assumption was satisfied based on the global result.

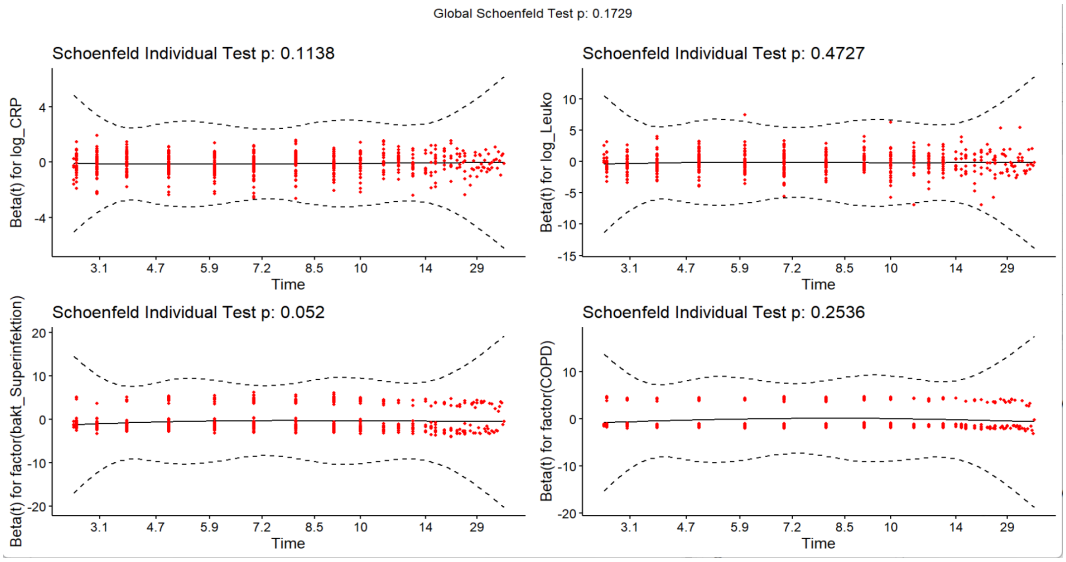

Figure 17: Schoenfeld test result for Influenza A

For RSV patients, hospital stays served as the dependent variable, and nosocomial infection, diabetes, bacterial infection were the independent variables. The results of the Schoenfeld test are as Figure 18. In Cox proportional hazards models, determining whether the overall model satisfies the proportional hazards assumption usually depends on the "GLOBAL" statistics, not just the results of the individual variables.

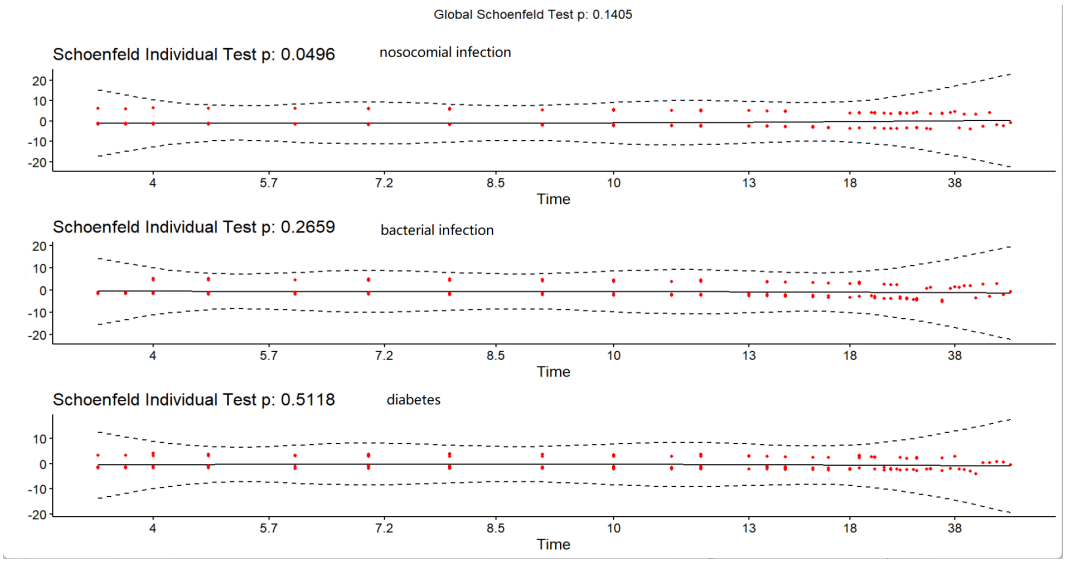

Figure 18: Schoenfeld test result for RSV

For Influenza B patients, hospital stays served as the dependent variable, and bacterial infection, nosocomial infection, and diabetes were the independent variables. The results of the Schoenfeld

test are as Figure 19. The proportional hazards assumption was satisfied based on the global result.

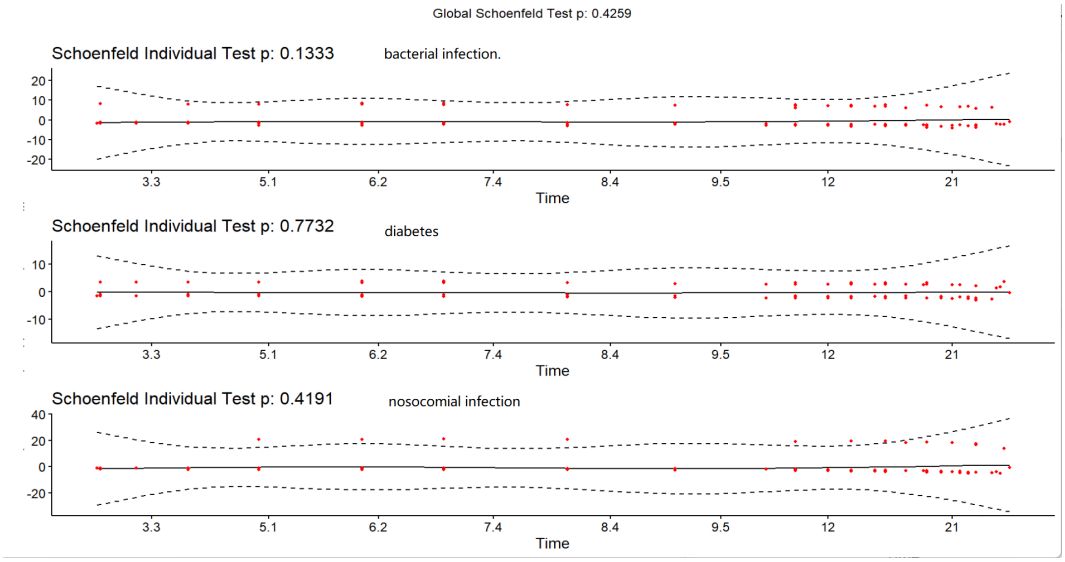

Figure 19: Schoenfeld test result for Influenza B

For SARS-CoV-2 patients, hospital stays served as the dependent variable, and bacterial infection, nosocomial infection, and log(CRP) were the independent variables. The results of the Schoenfeld test are as Figure 20. The proportional hazards assumption was satisfied based on the global result.

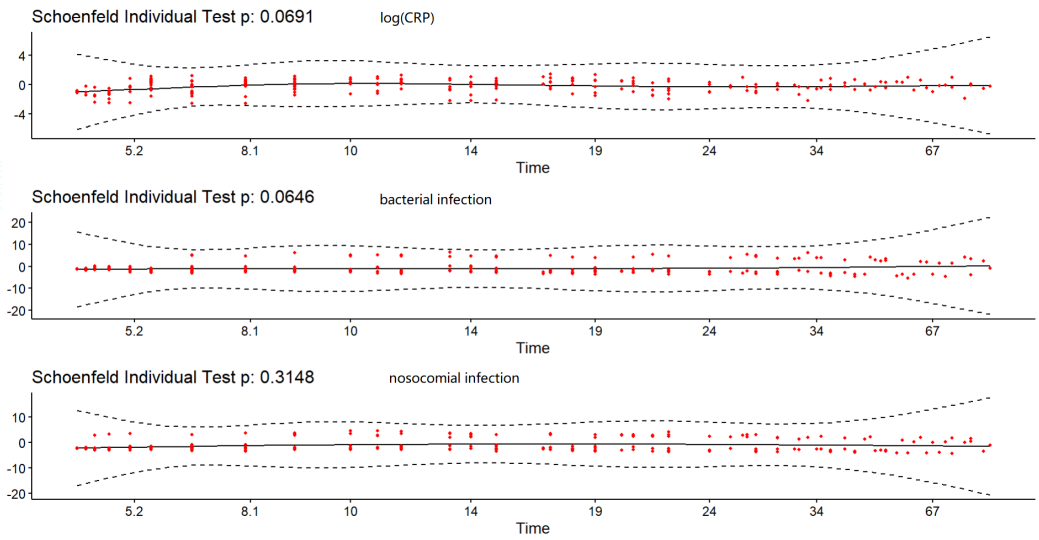

Figure 20: Schoenfeld test result for SARS-CoV-2

For Omicron patients, hospital stays served as the dependent variable, and age, diabetes, and log(LDH) were the independent variables. The results of the Schoenfeld test are as Figure 21. The proportional hazards assumption was satisfied based on the global result.

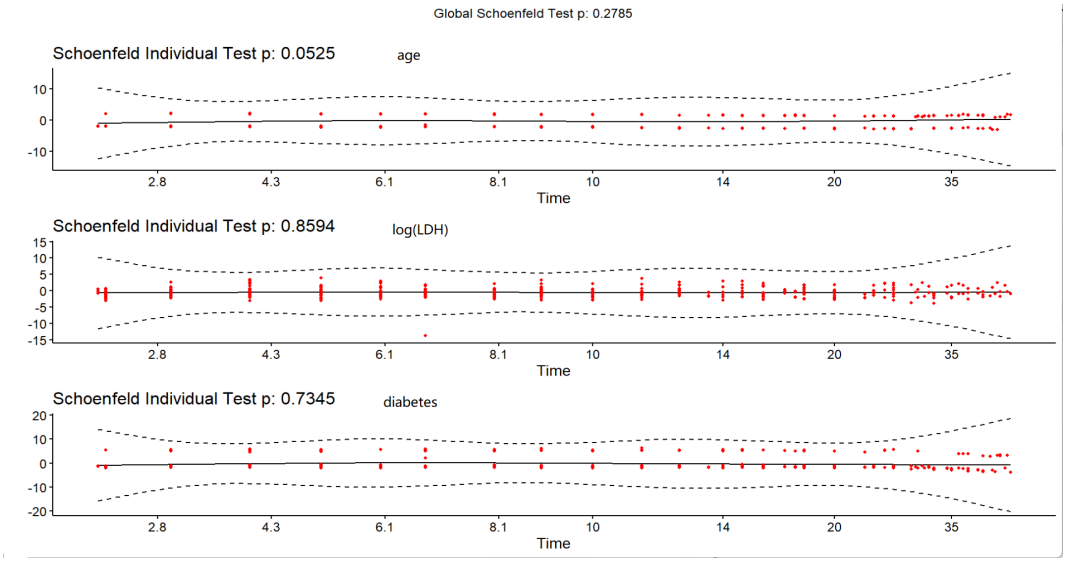

Figure 21: Schoenfeld test result for Omicron

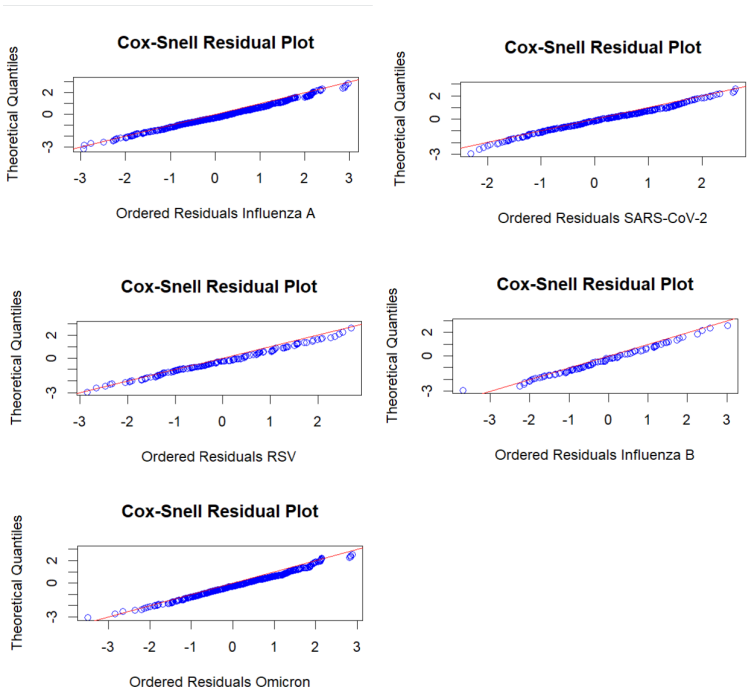

Figure 22: Cox-Snell residuals

The Cox-Snell plots were used to assess the proportional hazards model for the five respiratory infections (Figure 22). The residuals of the five plots showed less deviation from a straight line through the origin with a slope of one. However, it should be noted that all five plots show the same state, namely that the residuals are unevenly distributed in each of the beginning and final segments.

3.3.2 Independence Assumption

The cumulative risk curves of the actual observed values closely align with the expected cumulative risk curves in all five plots. The denser distribution of points above the x-axis and the looser distribution of points below the x-axis in the plot indicates that the model tended to overestimate the cumulative risk at higher survival times and underestimate it at lower survival times.

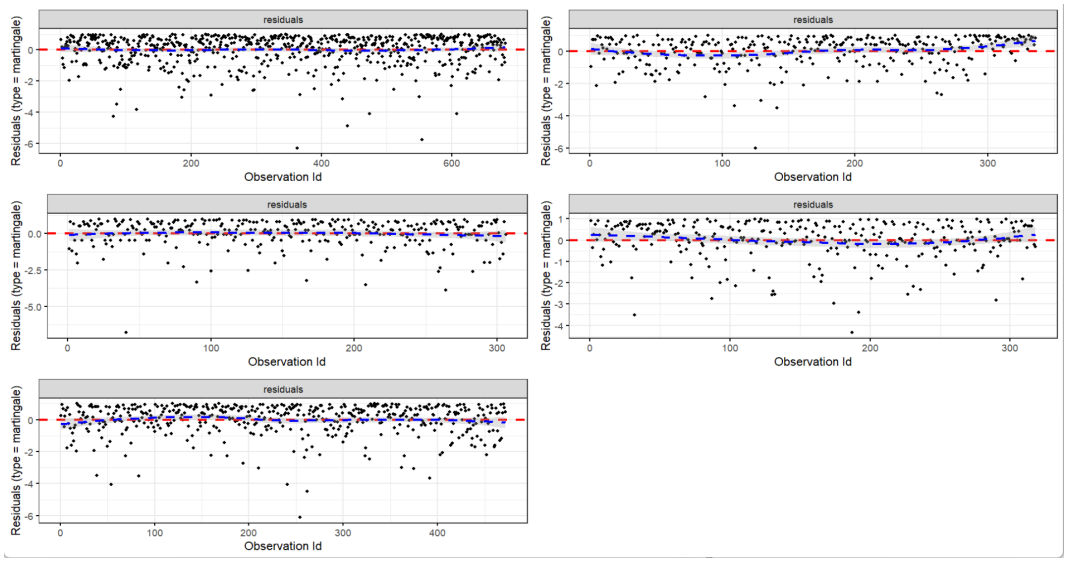

Figure 23: martingale residuals, from left to right, top to bottom, Influenza A, RSV, Influenza B, SARS-CoV-2 and Omicron

3.4 Comparison of SARS-CoV-2 and Omicron by RMST

Factors that showed a statistically significant effect on hospital stays by using Cox regression were analyzed again using the RMST method. RMST analysis emphasizes group comparisons to investigate the effect of hospital stays. Due to the involvement of medical knowledge, there is no basis for stratifying the laboratory values in this research. Consequently, the impact of laboratory values was not discussed using this method.

The average survival time was 12.32 days in the Omicron group and 17.66 days in the SARS-CoV-2 group. The Omicron group had 472 patient data after data cleaning, of which 430 patients had a length of stay less than or equal to 30 days. For the SARS-CoV-2 group, 273 patients out of 318 data had a length of stay less than or equal to 30 days. From Figure 4, it can be seen that there was a longer survival time in the SARS-CoV-2 group, i.e., there was a right-skewed tendency, RMST method was employed to focus on events occurring within 30 days.
All the following analyses were performed on the dataset after multiple imputations.

3.4.1 Nosocomial infection

The restricted mean survival time of up to 30 days for SARS-CoV-2 patients with nosocomial infection was about 21.78 days, and without nosocomial infection was about 15.12 days (Figure 24). This difference is statistically significant and showed that patients with nosocomial infection had a longer hospital stay compared to patients without nosocomial infection. In the Omicron group, the impact of nosocomial infections on hospital stay was more significant, causing a significant time difference.
For Omicron data after multiple imputations, the longest hospital stay for a patient with nosocomial infection is 55 days. The general trend in impact was consistent with the SARS-CoV-2 group, being that patients with nosocomial infection had longer hospital stays. The RMST for patients with nosocomial infections was approximately 23.29 days, compared with approximately 9.52 days for those with non-nosocomial infections (Figure 25), the impact of nosocomial infections on hospital stay was more significant than SARS-CoV-2 group, causing a significant time difference.
The intensity of the effect of these factors on the hospital stay of both groups was compared by comparing RMST (arm=1)/(arm=0). Patients with nosocomial infections and Omicron had, on average, 1.44 times longer hospital stays than patients without nosocomial infections, and 2.44 times for SARS-CoV-2 patients. There was less difference in RMST between the two groups compared to the Omicron group.

3.4.2 Pre-existing conditions, bacterial infection

The results of RMST of bacterial infection and pre-existing conditions for SARS-CoV-2 patients and Omicron patients are shown in Table 6. The effect of these factors on hospital stays was significant, with patients with these conditions (arm=1) having longer mean hospital stay within the tau (30 days) compared to patients without the specific factors (arm=0).
Table 7 shows in these four factors that bacterial infections bring about more significant differences, especially for the SARS-CoV-2 group. The trend in the intensity of this effect was the

same as that derived from the Cox regression.

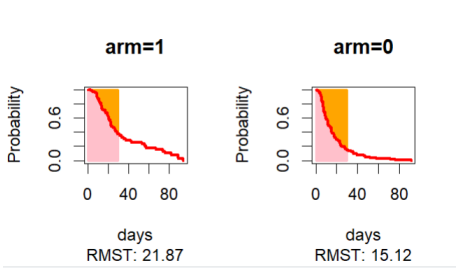

Figure 24: The results of the SARS-CoV-2 by RMST: the impact of nosocomial infection (arm=1) on the hospital stays compared to non-nosocomial infection (arm=0).

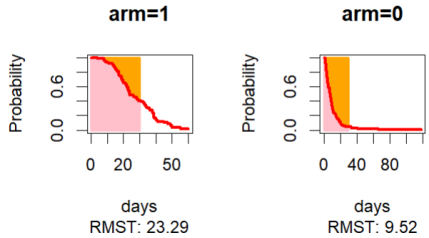

Figure 25: The results of Omicron by RMST: the impact of nosocomial infection (arm=1) on the hospital stays compared to non-nosocomial infection (arm=0).

	SARS-CoV-2	RMST (1)	RMST (0)	Omicron	RMST(1)	RMST (0)
bacterial infection		24.014	15.195		14.110	11.387
Diabetes Mellitus		19.686	16.489		13.019	11.288
Cardiac problem		20.236	15.147		12.650	10.606
DRI		21.306	16.138		13.625	11.046

Table 6: The results of two groups obtained by using RMST method: tau= 30, 1 means the situation occurred, 0 means it did not

	SARS-CoV-2	(arm=1)/(arm=0)	Omicron	(arm=1)/(arm=0)
bacterial infection		1.58		1.23
Diabetes Mellitus		1.19		1.15
Cardiac problem		1.33		1.19
DRI		1.32		1.23

Table 7: RMST (arm=1)/(arm=0) in SARS-CoV-2 and Omicron group

4 Discussion

To investigate the effect of different variables on the survival time of patients, most researchers often use proportional hazard modeling. However, Cox regression analysis should be noted that the proportional hazard assumption is satisfied, and if this assumption is not satisfied, the model results may be misleading[16]. RMST has been promoted as an alternative and better summary statistic for survival data when proportional hazard assumption is violated[17].
Cox regression is indeed concerned with the hazard or risk of an event occurring at a given time, which is related to the probability of the event happening within a small time interval. A Cox model estimates hazard ratios, which measure how much a covariate affects the hazard function for the event of interest[18]. For example, the HR for the older than 76 group is equal to 0.76, which means that the hazard of discharge in the older group is 24% lower than the hazard in the younger group, but it does not imply that the hospital stays decreased to a certain percentage (Figure 26-27). On the other hand, Restricted Mean Survival Time (RMST) is a measure of the average survival time up to a certain time point. It estimates the average time until an event occurs (Figure 24-25).

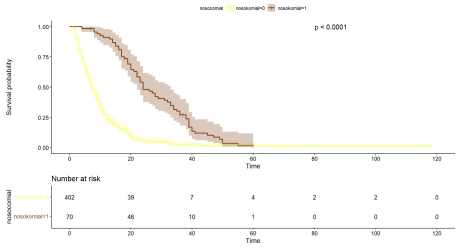

Figure 26: Omikron: upper curve is for with nosocomial infections

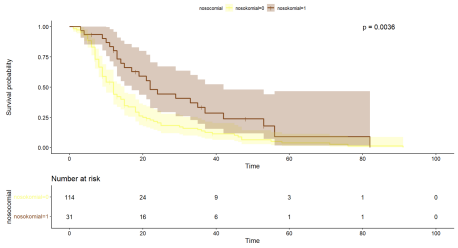

Figure 27: Sars-Cov-2: upper curve is for with nosocomial infections

In all five groups of patients normally discharged from the hospital, there were cases, where hospital stays exceeded 100 days (Figure 4), yet the average stays in each group were less than 20 days. In Cox regression, variations in the range and distribution of survival times in the dataset may affect the results and interpretation of the model. Some extreme values of survival time may have an impact on the estimation of Cox regression[19]. RMST is used to measure the average survival time within a specific cutoff time tau, and can effectively mitigate the influence of extreme values in survival times.
Existing methods for regression modeling of RMST do not apply to left-truncated right-censored data that arise frequently in prevalent cohort studies[20]. Left-truncated right-censored refers to the fact that some individuals in the study were present at the beginning of the observation period (i.e., left-truncated) but remained event-free at the end of the observation period (i.e., right-censored). Because the observation period did not cover the entire survival time of these

individuals, it is not possible to know exactly how long they survived. In this thesis, this is reflected in the fact that it is uncertain whether each patient was admitted to the hospital on the first day of the disease.
Cox regression assumes that the risk ratio is constant, can manage censored data, and has no specific requirements for data distribution. RMST is estimable under heavy censoring[21].
In conclusion, there are no existing standards that can directly determine the best method for survival analysis, but only the most appropriate method based on the dataset. In general, when the proportional hazard assumption is satisfied, the test based on the RMST is almost as effective as the test based on the COX regression. When the proportional hazard assumption is not satisfied, the test based on the RMST is better[22].

5 Conclusions

Regardless of the virus type, analysis using Cox regression analysis showed that increasing age, increasing laboratory values (CRP, glucose, leukocyte), having pre-existing conditions, having nosocomial infections, and having bacterial infections all increased hospital stays. In contrast, an increase in the laboratory value of LDH decreased hospital stays. The same results were obtained when analyzing Omicron and SARS-CoV-2 using the RMST method: increasing age, increasing laboratory values (CRP, glucose, leukocyte), pre-existing conditions, nosocomial infections, and bacterial infections prolonged the hospital stays.

Among the many factors, the effects of the presence of nosocomial infections and the presence of bacterial infections were the most significant, with HRs of less than 0.6. And the effect of increased blood glucose levels on hospital stays in SARS-CoV-2 patients was also significant (HR=0.41).

Age and nosocomial infection had the greatest impact on Omicron patients. Pre-existing conditions (diabetes, DRI, heart problems) and bacterial infection affected SARS-CoV-2 patients the most.

The graphical representation of the correlations and Goodness-of-Fit tests of multivariate models after multiple imputations provides a foundation for future multivariate analysis as well as the construction of predictive models.

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